Entity Resolver: Solution Architecture

DSAN 6700 Final Project, Deliverable 2 (Midpoint Check-in)

Team: [team member names]

Application on Jetstream2: http://[instance public IP]:8000/ · GitHub: **https://github.com/normangw/entity-resolver**

Problem. The same company appears under many names across datasets ("Wal-Mart Stores Inc.", "沃尔玛", "Walmart"), so tables cannot be joined or deduplicated. The service maps a messy or non-English company name, or a whole spreadsheet column of them, to one canonical company with its Wikidata ID, parent and headquarters, and says plainly when it is not sure.

Process: use cases

APPLICATION BUILD

- Build the alias index from Wikidata (`scripts/build_index.py`): about 14k listed companies with multilingual aliases
- Build the labeled evaluation set (`make_eval_set.py`, 176 queries)
- Package as a Docker image; deploy with Compose

APPLICATION USE

- Type one name, pick region, see the canonical company, confidence and source
- Upload an .xlsx / .xls / .csv, pick the sheet and the company-name column, pick a country or region
- Review summary and integrity checks, download the file with the original rows plus resolver columns
- Call `POST /resolve` directly from code

APPLICATION MONITORING

- `/health` (process up) and `/ready` (index loaded)
- Docker health check; `restart: unless-stopped`
- `scripts/evaluate.py` accuracy report; `scripts/bench.py` latency

Execution: API and functionality

FRONTEND

- One static page served by the API at `GET /`: name form with example chips, and a spreadsheet panel with sheet, header, column and region pickers, a preview table, progress, checks and download

EXPOSED API (FASTAPI, DOCUMENTED AT `/DOCS`)

- `GET /health`, `GET /ready`
- `POST /resolve`
- `POST /upload`, `GET /upload/{id}/preview`, `POST /upload/{id}/resolve`, `GET /upload/{id}/status`, `GET /upload/{id}/download`

BACKEND LOGIC (NOT EXPOSED)

- Normalize: Unicode NFKC, accent folding, lowercase, legal-suffix removal
- Score with RapidFuzz `WRatio` against all aliases, limited to the chosen region; below threshold 0.8 the answer is `unresolved`
- Low-confidence matches get a GLEIF legal-name check (budgeted, off by default)
- Upload jobs: safe file reading, one lookup per unique name, write result, read it back and verify

BASELINE AND MEASUREMENT

- Baseline: exact match on the normalized alias, 79.9% accuracy on 176 labeled queries; the fuzzy scorer reaches 95.8% (`eval/README.md`)

State: what persists, and where

SOURCE CONTROL

- GitHub repo, `main` plus feature branches merged by pull request; GitHub Actions CI runs tests and lint

DATA STORAGE

- Alias index `companies.json`: versioned with the code, copied into the named Docker volume `alias-index`
- No database and no user accounts. Uploaded files and results sit in a temporary directory and are deleted one hour after upload; cell contents are never logged

MODELS AND CONFIGURATION

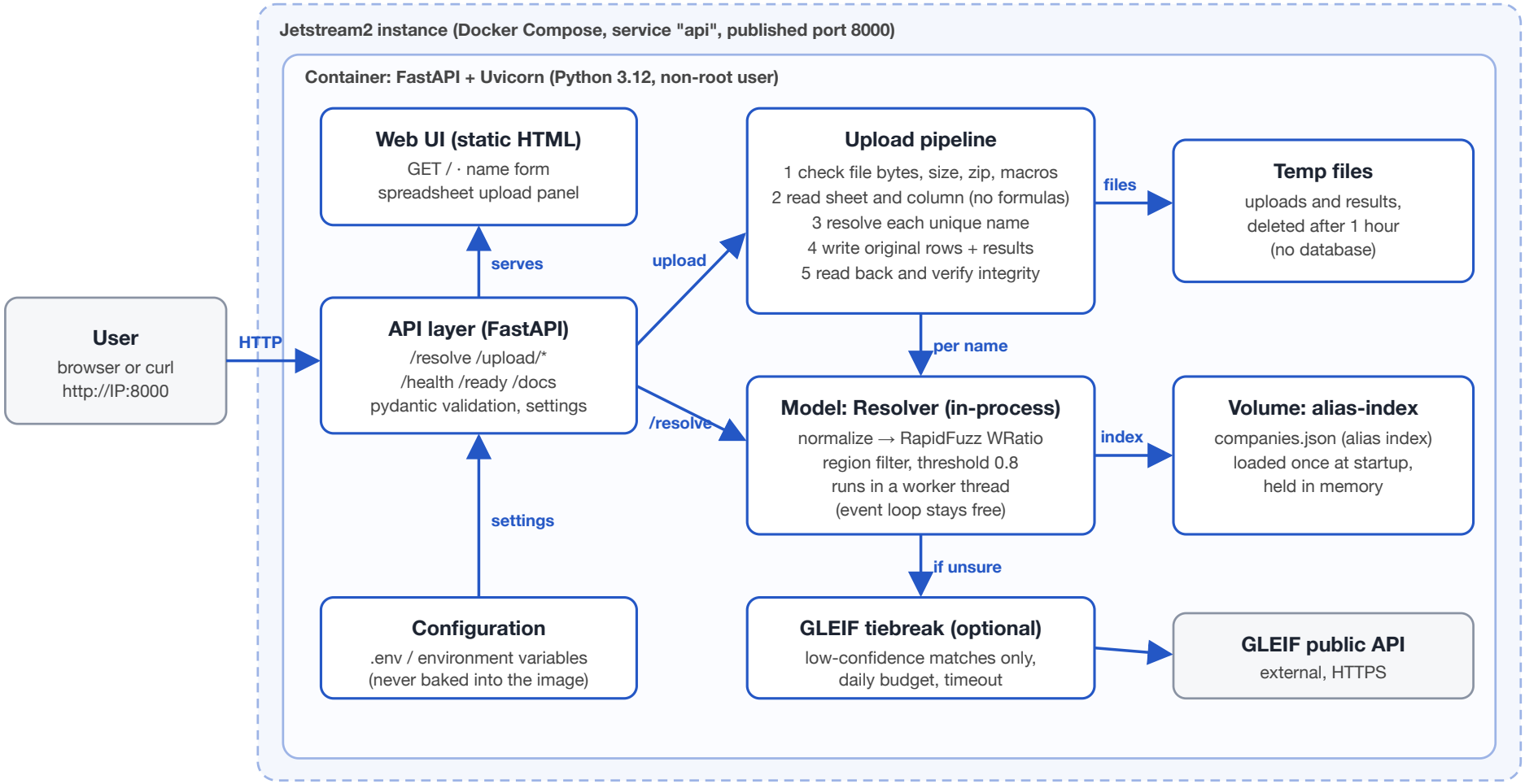
- The "model" is the alias index plus the scorer settings; update by editing the index and rebuilding the image, roll back with a git revert and rebuild
- All settings come from environment variables (`.env`, never baked into the image): threshold, GLEIF switch, upload limits

TEMPORARY AND CACHED

- Alias index held in memory after startup
- In-memory table of upload jobs and progress
- GLEIF daily request counter (in memory)

Entity Resolver: Technical Architecture

Programmatic flow between components, deployed with Docker and Compose on a Jetstream2 instance. Only port 8000 is published.



Layer	Technology	Notes
Frontend	One static HTML page served by the API	No separate app; loads nothing from the internet
API	FastAPI, Uvicorn, pydantic, python-multipart	OpenAPI docs at `/docs`; startup load is non-blocking, so `/health` answers while `/ready` is 503
Model service	RapidFuzz scorer in the API process	About 1.3 ms per name, no GPU, so a second container would only add a network hop
Storage	JSON alias index in a Docker volume; temp directory for uploads	No database or vector store yet, so there is no second port to protect
Deployment	Docker image (uv, non-root), Compose, Jetstream2	Only 8000 is published; `restart: unless-stopped`